

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of M. Pharm

(Pharmaceutical Technology and Pharmaceutics (Drug Regulatory Affairs))

University Theory Examination January 2018

PHCC001 BASIC ANALYTICAL TECHNIQUES

Date: 01.01.2018 , Monday

Time: 10.00 a.m. to 01.00 p.m

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1 A) Write a note on elution principle for column and planar chromatography. 05
B) Classify stationary phase used in Gas chromatography (GC) and explain the factors governing the selection of stationary phase in GC. 05
- Q.2 A) Enlist various factors affecting separation in HPLC and describe each in brief. 05
B) Discuss mobile phase selection and optimization for HPTLC. 05
- OR
- Q.2 A) Write in brief about theory and applications of TGA. 05
B) Discuss the principle of affinity and size exclusion chromatography. 05
- Q.3 Answer the following (Any three) 15
(A) Explain validation of an analytical method.
(B) Write a note on detectors used in HPLC
(C) Differentiate between DSC and DTA.
(D) Differentiate between GC and HPLC.

SECTION-II

- Q.4 (A) Discuss the detectors used in UV-Visible spectrophotometer. 05
(B) Enlist different mass analysers. Write note on quadrupole mass analyser. 05
- Q.5 (A) Discuss sample preparation methods used in IR. 05
(B) Explain Bragg's law. Write applications of X-ray diffraction technique. 05
- OR
- Q.5 Write note on Proton NMR technique. 10
- Q.6 Answer the following (Any three) 15
(A) Write note on FT-IR Technique.
(B) What is electrophoresis? Classify electrophoretic methods and discuss its basic principle.
(C) What are related substances and impurities? Discuss different techniques to identify impurities.
(D) Discuss the method of preparation and characterization of reference standard compound.
